



Research paper

A head-to-head multi-parametric high content analysis of a series of medium chain fatty acid intestinal permeation enhancers in Caco-2 cells



David J. Brayden*, John Gleeson, Edwin G. Walsh

School of Veterinary Medicine and Conway Institute, University College Dublin, Dublin, Ireland

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ABSTRACT

There is debate over the narrow safety margin of many oral permeation enhancers. The sodium salts of medium chain fatty acids (MCFAs) are components of selected oral peptide formulations being assessed in advanced clinical studies. The aim of the study was to examine the effects of the C₈–C₁₂ series on filter-grown Caco-2 monolayers and cells grown on 96 well plates in order to dissect the relationship between paracellular permeability enhancement (Papp of [¹⁴C]mannitol, reduction in TEER), critical micellar concentration (CMC), and sub-lethal cytotoxicity (high content analysis (HCA)). There was a high degree of correlation between the EC₅₀ for increasing the Papp with reduction in transepithelial electrical resistance (TEER), and with CMC values. C₈ had the highest EC₅₀ and highest CMC value and was the least cytotoxic, while C₁₂ had the reverse, suggesting a close association between increases in chain length with increases in permeability, hydrophobicity and cytotoxicity. HCA revealed further association between MCFA-induced intracellular calcium increases and plasma membrane permeability reductions in Caco-2 cells with the EC₅₀ to increase the Papp across monolayers. HCA identified sub-lethal cytotoxicity in a series of MCFA and related cell parameters to physicochemical properties and efficacy as intestinal permeation enhancers. These mild surfactants therefore non-specifically partition into the plasma membrane causing membrane fluidization, which is associated with concentration-dependent increases in permeability.

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1. Introduction

High content analysis (HCA) was originally used by the pharmaceutical industry to test preclinical toxicity of discovery molecules in liver cells and to identify associated structure-activity-relationships. It determines subtle changes in cell health with the aim of reducing toxicity-associated attrition during late stage drug development [1]. Based on fluorescent microscopy of live single cells in a population, it is a powerful qualitative and quantitative tool to con-

temporarily measure multiple cellular parameters in response to challenge over a set of time points and concentrations. Potential molecule toxicity is multi-factorial and depends on cell homeostasis [2]. No single cell parameter will definitively detect adverse effects and accurately predict toxicity potential *in vivo*. Unlike conventional cytotoxicity assays including the 3-(4,5-dimethylthiazol-2-yl)-2,5 diphenyltetrazolium bromide (MTT) conversion assay, which detects gross cytotoxicity, HCA provides subtle pre-lethal screening that has high sensitivity for detecting potential human toxicity and high specificity for a wide range of cellular effects [3]. Metrics are read in individual cells by combining multi-well formats and sensitive fluorescent dyes, with the use of inverted fluorescent microscopy and software to quantify changes in fluorophore staining. In the first report of HCA in a drug delivery study, the sensitivity of HCA was an order of magnitude greater than that of the MTT assay in assessing poly((2-dimethylamino)ethyl methacrylate) (pDMAEMA) cytotoxicity in Caco-2 cells and U937 monocytes, with the Caco-2 cells proving far more resilient than monocytes [4].

An ideal intestinal permeation enhancer (PE) for peptides, proteins and macromolecules should reversibly increase epithelial flux

Abbreviations: CMC, critical micelle concentration; HCA, high content analysis; HR-US, high resolution ultrasonic spectroscopy; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5 diphenyltetrazolium bromide; MCFAs, medium chain fatty acids from C₈–C₁₂; C₈, sodium salt of caprylic acid; C₉, sodium salt of nonanoic acid; C₁₀, sodium salt of capric acid; C₁₁, sodium salt of undecanoic acid; C₁₂, sodium salt of dodecanoic acid; PE, permeation enhancer; Papp, apparent permeability coefficient; EC₅₀, concentration of MCFA giving a half maximal Papp; CN, cell number; NA, nuclear area; NI, nuclear intensity; IC, intracellular calcium; MMP, mitochondrial membrane potential; PMP, plasma membrane permeability (PMP).

* Corresponding author. Room 213, School of Veterinary Medicine and The Conway Institute, University College Dublin, Belfield, Dublin 4, Ireland. Tel.: +353 1 7166013; fax: +353 1 7166204.

E-mail address: david.brayden@ucd.ie (D.J. Brayden).

without local or systemic toxicity. PEs are essential components of most oral peptide dosage forms in clinical trials including acyl carnitines [5], 8-(N-2-hydroxy-5-chloro-benzoyl)-amino-caprylic acid [6], sodium caprylate (C_8) [7] and sodium caprate (C_{10}) [8]. Specifically in respect of medium chain fatty acids (MCFAs) as PE, clinical data from Merriam Pharmaceutical's (Ireland) GIPETTM matrix tablet containing C_{10} have yielded significant oral bioavailability of poorly permeable molecules with very low numbers of adverse events in several hundred volunteers in multiple Phase 1 studies [9]. The "Transient Permeability Enhancer" technology of Chiasma (Israel) uses C_8 in solid hydrophilic peptide-containing particles mixed with a hydrophobic medium to give an oily suspension, and their enteric-coated capsule of octreotide has been advanced in trials for acromegaly patients [10]. Even though MCFAs are found in many foodstuffs and have a long history of use in man, there is a perception of a safety issue in the event of repeated oral dosing of these surfactants, due to epithelial abrasion and potential absorption of bystander pathogens, toxins and lipopolysaccharide fragments [11,12].

In respect of *in vitro* effects of MCFA, Lindmark et al. demonstrated that while both C_{10} and C_{12} inhibited phospholipase C (PLC), only C_{10} redistributed tight junction (TJ) proteins [13], and the same group demonstrated that for the even-numbered MCFA, the rank order of efficacy was $C_{12} > C_{10} > C_8 > C_6$ in respect of inducing paracellular [^{14}C]-mannitol fluxes across Caco-2 monolayers [14]. In contrast, others showed that while C_{10} permitted increased paracellular flux of solutes with molecular weights (MW) < 10 kDa in HT29/B6 monolayers [15], C_{12} enabled flux of molecules of MW < 500 Da [16]. Therefore, rank ordering of the efficacy of MCFA as PEs seems to depend in part on the *in vitro* bioassay model and parameters being assessed. Moreover, while there is consensus that MCFA enhance flux of hydrophilic molecules by opening tight junctions arising from complex intracellular mechanisms initiated at the plasma membrane *in vitro*, there is less certainty on whether this mechanism can solely account for *in vivo* oral bioavailability data achieved with extremely high doses of the MCFA.

Our objective was to use HCA to produce mechanistic data on epithelial cellular responses associated with C_8 – C_{12} , and to identify a relationship between permeation enhancement capacity and sub-lethal cytotoxicity and other physicochemical parameters. The series (including the odd numbered MCFA that do not naturally occur in humans) has never been examined under such multiple criteria in a systematic way. The data show clear relationships between the physicochemical properties of the MCFA series (critical micellar concentration (CMC) data), their permeation enhancing activity across Caco-2 monolayers (mannitol apparent permeability coefficient (Papp) data), and cytotoxic potential (MTT and HCA data). HCA indirectly supports a key role of fatty acid insertion into epithelial plasma membranes as a primary event in the mechanism of action of MCFA as PE.

2. Materials and methods

2.1. Materials

The sodium salts of caprylic (C_8) and dodecanoic (C_{12}) acid were obtained from Sigma Aldrich (Ireland). The sodium salt of capric acid (C_{10}) was obtained from Fluka. Nonanoic and undecanoic acids were also supplied by Sigma Aldrich; their respective C_9 and C_{11} sodium salts were made by mixing the acids with NaOH at a 1:1 M ratio in deionised water at 50 °C with 1% isopropyl alcohol. Salt solutions were oven-dried overnight at 65 °C to form powders, which were tested by differential scanning calorimetry and thermo-gravimetric analysis. Purity was >95% for all MCFA except

C_9 , which was 93% due to some unreacted nonanoic acid (data not shown).

2.2. Critical micelle concentration (CMC)

High resolution ultrasonic spectroscopy (HR-US) was used to measure CMC of solutions in Ca^{2+} -free DMEM [17]. Ultrasonic wave velocity (v (m/s)) was measured between 5 and 15 kHz using a HR-US 102 instrument (Ultrasonic Scientific Ltd., Ireland). MCFA stock solutions were prepared in de-gassed Ca^{2+} -free DMEM. 1 ml de-gassed medium was added to the cell and the base v value determined at 37 °C. Stock MCFA solutions were then centrifuged (>10,000g) to ensure that they were completely de-gassed. v was measured while the sample was titrated by adding MCFA stock solution through a septum in the lid of the measuring cell using a manual Hamilton dispenser and 250 μ l Hamilton syringe.

2.3. Fluxes across Caco-2 monolayers

Caco-2 cells (ECACC, passages 40–50) were cultured on plastic until confluent and then on 12 mm polyester Transwells[®] for 21–28 days [18]. Transepithelial electrical resistance (TEER) and [^{14}C]-mannitol fluxes across Caco-2 monolayers were measured over 180 min and the apparent permeability coefficient (Papp) of mannitol was calculated [18].

2.4. MTT cytotoxicity assay

Caco-2 cells were cultured on 96 well plates in DMEM, 5% foetal bovine serum and 1% minimal essential amino acids. The assay was carried out at 1 and 8 h exposure times according to previous descriptions [19]. At 24 h, DMEM was replaced with Ca^{2+} -free DMEM in order to avoid precipitation of MCFA salts [20]. The IC₅₀ was defined as the concentration of MCFA required to reduce the absorbance of the MTT-formazan crystals by 50%.

2.5. HCA in Caco-2 and HepG2 cells: 96 well plates

The HCA protocol [3] was adapted for Caco-2 cells, which were grown as above. HepG2 cells were used in the study as a HCA reference for systemic exposure of liver cells over 72 h [3]. A total of 6000 Caco-2 cells were seeded in a volume of 200 μ l DMEM/well in each well of Nunclon[®] 96 well plates at room temperature for 60 min prior to 24 h incubation at 37 °C. After 24 h, medium was replaced with 300 μ l Ca^{2+} -free DMEM containing MCFAs and cells were exposed throughout for 1 and 8 h for Caco-2 cells, and at 72 h for HepG2. 150 μ l medium was then removed and a dye mix (150 μ l of 2 $\times$ stock in Ca^{2+} -free medium) was added. It contained final concentrations of 0.8 μ M Hoechst 33342 (to measure cell number (CN), nuclear area (NA) and nuclear intensity (NI); 1 μ M Fluo-4 AM (to measure intracellular calcium (IC); 20 nM tetramethyl rhodamine methyl ester (TMRM) (to measure mitochondrial membrane potential (MMP); and 1 μ M TOTO-3 iodide 642/660 (to measure plasma membrane permeability (PMP). Cells were incubated with the dyes for 50 min before positive controls were added for each non-nucleus parameter: 100 μ M carbonylcyanide-*p*-trifluoromethoxyphenylhydrazone (FCCP) for MMP, 20 μ M ionomycin for IC, and 0.05% (v/v) Triton[®] X-100 for PMP. 50 μ l medium was removed from each positive control well and replaced with 50 μ l of the positive control agent. The plate was incubated for a further 10 min before image acquisition using an In-Cell[®] 1000 High Content Analyzer (GE Healthcare, UK) using multi-target analysis. The excitation/emission wavelengths used were Hoechst 33342: 360/460 nm, Fluo 4-AM: 480/535 nm, TMRM: 535/600 nm and TOTO-3: 620/700 nm. The Hoechst channel was used for cell counts and 10 random fields per well were imaged

using a 10× objective magnification. For each other parameter, 8 random fields per well were imaged using a 20× objective magnification. Optimal exposure times were for 150 ms (Hoechst 33342), 500 ms for Fluo-4 AM, 600 ms for TMRM and 200 ms for TOTO[®]-3 iodide.

2.6. HCA on Caco-2 cells: Transwells[®]

The response of differentiated Caco-2 monolayers to C₁₀ exposure for 60 min was evaluated. Cells were grown on Transwells[®] as above. Once TEER values of >400 Ω cm² were established, basolateral and apical-side media were removed and monolayers were washed three times with 0.5 ml PBS. Viable monolayers were needed to ensure accurate image acquisition and data analysis, as non-adhered dead cells bind dyes and multiple layers prevent imaging. 0.5 ml Ca²⁺-free DMEM was then added to the apical chamber and 1.5 ml DMEM added to the basolateral chamber and TEER was measured across monolayers to ensure integrity. 250 µl and 750 µl medium were removed respectively from the apical and basolateral chambers of each filter. 250 µl of Ca²⁺-free DMEM containing the same concentration dye mix as in Section 2.5 with or without C₁₀ was added apically to monolayers and 750 µl DMEM containing dye mix was added basolaterally. Plates were incubated for 60 min at 37 °C with 5% CO₂ in air. Similarly, 50 µl and 150 µl medium were removed from respective apical and basolateral chambers of positive control filters and replaced with an equal volume of 10× positive controls prepared in 1× dye mix in Ca²⁺-free DMEM. Transwell[®] filter dimensions were programmed into the In-Cell[®] 1000 and the monolayers imaged.

2.7. Statistical analysis

For HCA, MCFA treatments were added in duplicate to each plate of wells or filters and each experiment was run on 3 separate occasions. All data were normalised to the untreated control and comparisons were analysed by one-way ANOVA with Dunnett's post-test. For [¹⁴C]-mannitol P_{app} measurements across monolayers, treatments were added in duplicate to each of three plates and comparisons were determined by one-way ANOVA with Bonferroni's post-test to compare groups. Data are presented as the mean ± SEM with statistical significance indicated as **P* < 0.05, ***P* < 0.01 and ****P* < 0.001.

3. Results

3.1. CMC values decrease with increasing chain length

CMC values were measured for C₈–C₁₂ in Ca²⁺-free DMEM (Table 1). The values in Ca-free DMEM using the HR-US method were similar to those previously reported in HBSS for even-numbered MCFAs using surface tension [21]. As the chain length

increased, CMC values dramatically decreased due to the increased hydrophobicity of the alkyl chain.

3.2. Potency in increasing P_{app} of [¹⁴C]-mannitol across monolayers relates to chain length

The MCFA concentrations required to increase the P_{app} of [¹⁴C]-mannitol across monolayers decreased with increasing chain length, with minimum threshold concentrations of 45, 21, 7.5, 2.1, and 1.5 mM for C₈, C₉, C₁₀, C₁₁ and C₁₂ respectively. The basal P_{app} was ~2 × 10^{−7} cm s^{−1}, in accordance with literature values [18]. A P_{app} value of 1.0 × 10^{−5} cm s^{−1} was set as the maximum acceptable value for [¹⁴C]-mannitol across Caco-2 monolayers in the presence of MCFA, and is similar to the P_{app} seen with detergent controls in other studies [22]. Setting this as maximal efficacy, the concentration giving a half maximal P_{app} (EC₅₀) was estimated for each MCFA (Table 1). The most potent was C₁₂ (EC₅₀ = 1.8 mM) and the least potent was C₈ (EC₅₀ = 75 mM) and there was good correlation between chain length and potency. Details of the concentration effects of each MCFA on the P_{app} are provided (Suppl. Fig. 1).

3.3. TEER values decrease with increasing chain length

From the same studies in which we measured the P_{app} of [¹⁴C]-mannitol across monolayers in the presence of MCFA, we measured the TEER decreases and detected an inverse relationship between the EC₅₀ values for TEER reduction and carbon chain length, i.e. C₁₂ was again the most potent, followed by C₁₁, C₁₀, C₉ and C₈ (Fig. 1, Table 1). Importantly, only when the TEER decreased by >80% was significant corresponding increases in P_{app} values were then seen at these threshold MCFA concentrations. There was a correlation coefficient (*r*) of 0.98 between increasing chain length and the EC₅₀ for enhancing the P_{app} for [¹⁴C]-mannitol, and a value of 0.94 for the P_{app} increases versus TEER reductions (Table 1). The lowest *r* value was 0.67 for the relationship between chain length and TEER decrease and this is because it is a poorly discriminating parameter and displays wider variability. In summary, C₁₂ was the most potent MCFA at enhancing [¹⁴C]-mannitol flux and decreasing TEER across Caco-2 monolayers and it has the lowest CMC value.

3.4. Increasing the chain length increases cytotoxicity: MTT assay

MTT was carried out on Caco-2 cells exposed to the series for 1 and 8 h. Decreases in cell viability at both time points were clearly related to concentration for C₁₂, C₁₁, and C₁₀, whereas C₈ and C₉ caused few changes (Fig. 2). The most cytotoxic agents were C₁₁ and C₁₂, where respective IC₅₀ values of 5.0 and 8.5 mM were seen, even with 60 min exposure. While these values are not very high *per se*, it is notable that they are still within the range of concentrations that enabled an increased [¹⁴C]-mannitol P_{app} over 3 h.

For C₈–C₁₀, few cells were killed over a 60 min exposure, even at concentrations as high as 20 mM. The most potent MCFAs as PEs (i.e. C₁₁ and C₁₂) were therefore the ones that killed most cells at 1 and 8 h. To place these data in context, the most relevant exposure time for intestinal epithelia is likely to be 1–2 h, as these cells will only temporarily be exposed to oral formulations. The MTT assay therefore gave limited information at the 1 h time point, but did reveal cytotoxic effects of C₁₁ and C₁₂.

3.5. HCA of MCFAs on Caco-2 and HepG2 cells

Optimisation was then achieved in Caco-2 cells in two aspects in 72 h incubations (matching to the original design in HepG2 for 72 h [3]). First, we proved that studies should only be carried out

Table 1
MCFA effects on: CMC, [¹⁴C]-mannitol P_{app} and TEER across Caco-2 monolayers. All data is in mM.

A. MCFA	B. CMC in Ca ²⁺ -free DMEM	C. EC ₅₀ for [¹⁴ C]-mannitol P _{app} increase	D. IC ₅₀ for TEER reduction
C ₈	262.0	75.0	35.2
C ₉	108.3	35.0	30.0
C ₁₀	26.1	8.3	6.1
C ₁₁	6.2	4.0	2.5
C ₁₂	2.8	1.8	1.0

Correlation coefficients (*r*): A v B = 0.99; A v C = 0.98; A v D = 0.67; C v D = 0.94; B v D = 0.93; B v C = 0.74.

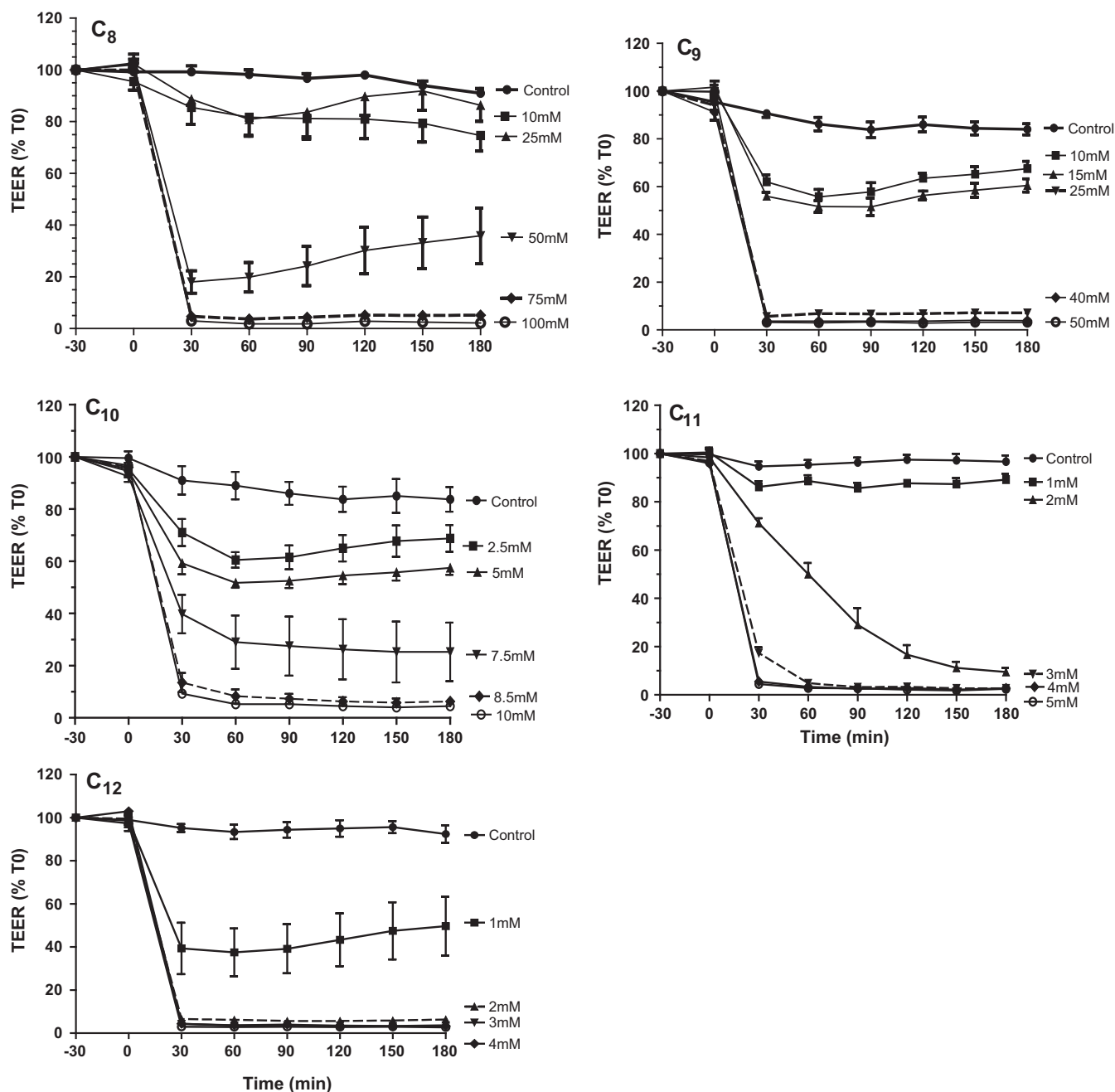


Fig. 1. Increasing MCFA chain length increases potency for TEER reductions across Caco-2 monolayers. The concentration that produced a significant increase in [¹⁴C]-mannitol P_{app} is highlighted in each case with a broken dashed line. Data for C₁₁ was reproduced from our previous study with permission from Springer Ltd. [37].

on the innermost 32 wells of each plate to cater for the edge effect in outer wells arising from evaporation leading to statistically reduced cell number (CN) and increased nuclear intensity (NI), IC and PMP compared to centre wells (Suppl. Fig. 2). Secondly, we confirmed that incubation in DMEM in the absence of calcium (required to avoid MCFA precipitation) retained Caco-2 parameters within ~10% of values in calcium-containing DMEM (Suppl. Fig. 3). A typical image analysis protocol is shown in an example with HepG2 cells (Fig. 3). We then confirmed that the positive control molecules for intracellular calcium (IC), mitochondrial membrane potential (MMP) and plasma membrane potential (PMP) caused dramatic 5, 2 and 2.5-fold changes in values in Caco-2 cells within 10 min (Fig. 4).

We then compared the MCFA series in calcium-free buffer after 60 min incubation with undifferentiated Caco-2 cells in respect of

the six parameters. Values are given as the minimum MCFA concentration that produced significant changes in each parameter compared to untreated controls (Table 2). A similar pattern of cellular responses was found for each agent. Initial effects were detected in the nucleus, with increased NI detected at concentrations of 30, 7.5, 2.5, 2.5 and 2 mM for C₈, C₉, C₁₀, C₁₁ and C₁₂ respectively. Fig. 5 shows fluorescence data for C₁₀ at different concentrations at 60 min. This was associated with increases in IC in a similar concentration and MCFA sequence. An exception was that C₁₂ caused a significant increase in IC at 1 mM compared to an increase in NI at 2 mM. Concentrations of MCFA altering IC also produced significant changes in MMP. The loss of MMP at 30, 12.5, and 2 mM C₈, C₉ and C₁₂ respectively was associated with very significant increases in IC. Less significant increases in IC induced by 5 mM C₁₀ were associated with mitochondrial

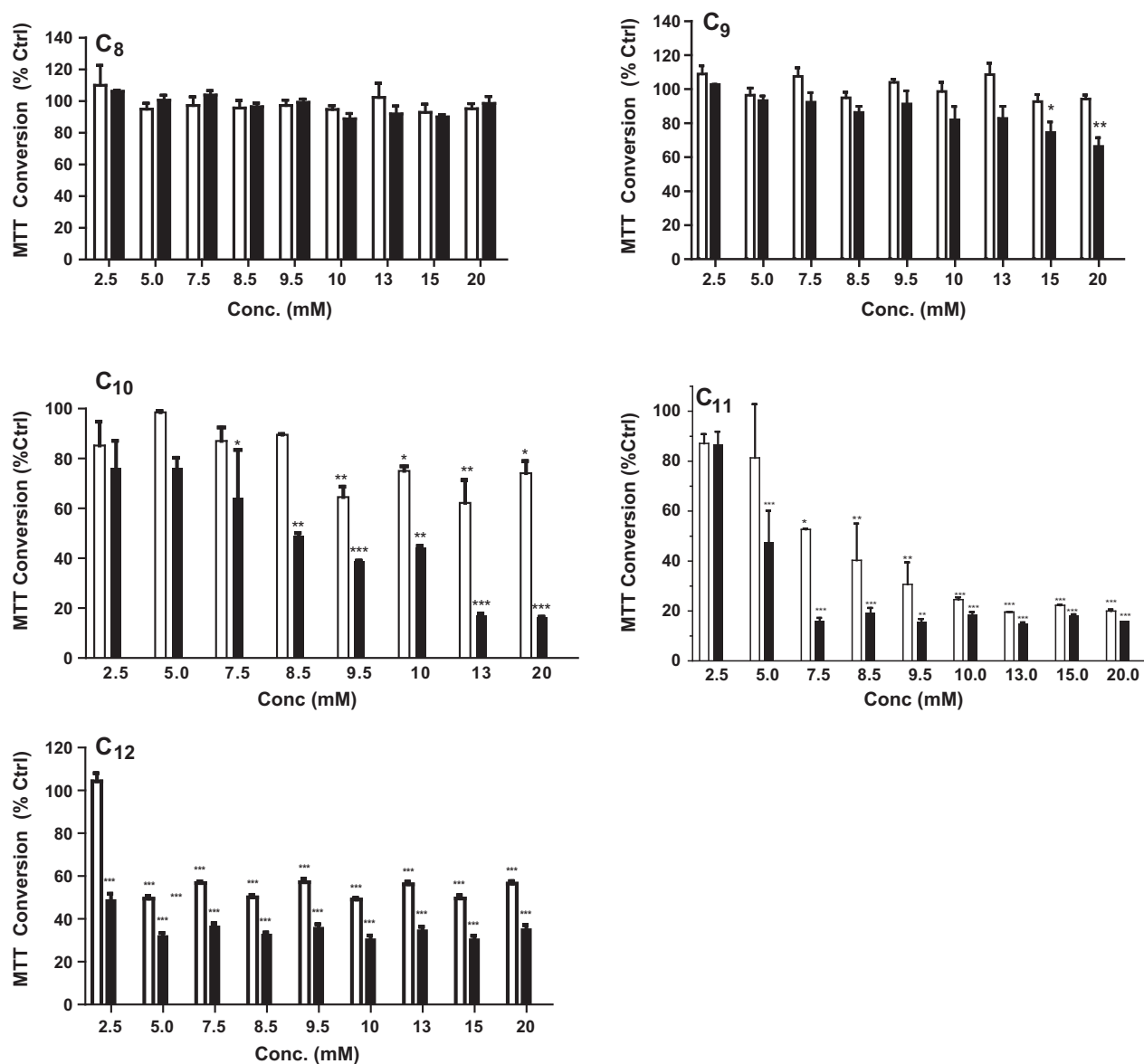


Fig. 2. MTT results for MCFA on Caco-2 cells grown on wells over time and concentration reveal a direct relationship with chain length. Results are presented as percentage of MTT converted compared to untreated media control (data not shown for 100% media control). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ relative to untreated controls, ($n = 3$). 1 h ■; 8 h □. Data for C₁₀ and C₁₁ was part reproduced from our previous study with permission from Springer Ltd. [37].

hyperpolarisation, a hormetic stress response to perturbation. No change in MMP was detected over a wide concentration range of C₁₁, whereas 20 mM C₁₀ significantly reduced it. Changes in cellular Ca²⁺ homeostasis and mitochondrial health were followed by increased PMP (40, 12.5, 8.5, 2.5, and 2 mM) for C₈–C₁₂ respectively. Threshold concentrations of 30 mM C₉, 13 mM C₁₁ and 2 mM C₁₂ caused a reduction in NA, indicative of nuclear fragmentation. 10 mM C₁₀ had the opposite effect, significantly increasing Caco-2 NA, likely to be due to hormesis. 40 mM C₈ significantly reduced CN, before reducing NA in surviving cells at a concentration of 75 mM. Cell numbers were reduced by 40 mM C₉ and 10 mM C₁₂, while neither C₁₀ nor C₁₁ reduced CN even at a concentration as high as 20 mM. Reductions in CN may be due to sloughing of cells from the well substrate and not cell death.

We then compared responses in Caco-2 and HepG2 cells to the MCFA over 72 h (Tables 3 and 4 respectively). This was done because 72 h was the time point of the original HCA assay in HepG2 cells for systemic exposure and a head-to-head comparison

between the lines is therefore relevant, but we recognise that the 60 min exposure time point is more important for intestinal epithelia. Increasing the exposure time to 72 h in Caco-2 cells resulted in significant reduction in CN at lower concentrations than seen at 60 min, particularly for the longer chain MCFAs. Importantly, MCFAs caused reductions in CN at concentrations 10–50% less than those causing increases in PMP. Similar to data from 60 min exposure, MCFAs induced a similar sequence of responses, with initial stress responses to perturbation manifested by increases in IC and NI, accompanied by mitochondrial hyperpolarisation. High IC was maintained resulting in Ca²⁺ overload and irreversible cell damage. In response to MCFA exposure, nuclear fragmentation, mitochondrial membrane collapse, and increased PMP were detected in Caco-2 and HepG2 cells *before* reductions in IC (Table 3).

HepG2 cells were more sensitive than Caco-2 cells, and this was particularly evident for C₁₁ and C₁₂. For example, 2.5 mM C₉ and C₁₀ equally reduced CN in Caco-2 and HepG2, while 10-fold and

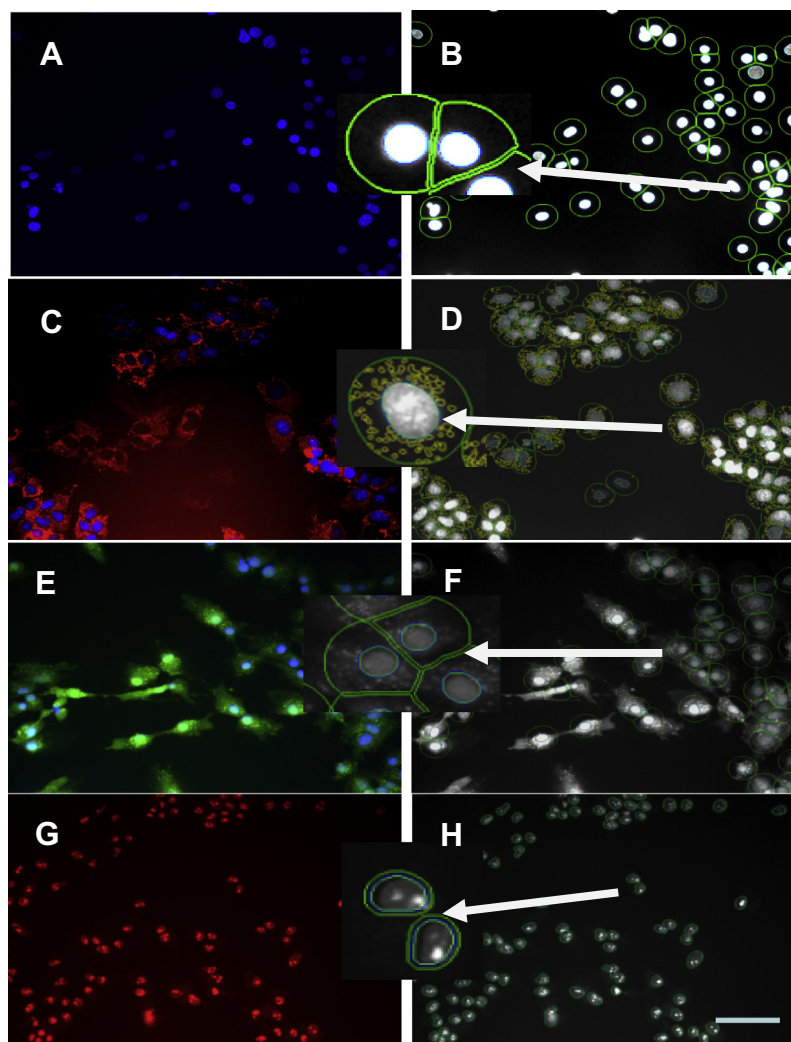


Fig. 3. Image analysis protocol. Images are from HepG2 cells for 72 h incubation. (A and B) Cells are identified by Hoechst 33342 staining of nuclei (blue). From staining analysis, the software provides information on cell number (CN), individual nuclear intensity (NI) and nuclear area (NA). (C and D) TMRM staining of mitochondria (red) on the basis of changes in mitochondrial membrane potentials (MMP). (E and F) Fluo-4-AM quantifies intracellular calcium (IC) in single cells (green). (G and H) TOTO-3 iodide 643/660 measures plasma membrane permeability (PMP) in single cells (magenta). Horizontal bar = 30 μm .

100-fold increases in C_{11} and C_{12} concentrations were required to reduce CN in Caco-2 compared to HepG2 respectively, (Table 4). This may reflect a capacity of enterocytes to resist perturbation by amphiphilic substances, with Caco-2 significantly more resistant to membrane perturbation induced by the more lipophilic C_{11} and C_{12} compared to HepG2. HCA shows that mM concentrations of C_8 – C_{10} are required to cause adverse effects to Caco-2 and HepG2 cells, even when exposure times are as high as 72 h. There would appear to be a chain length cut-off at C_{10} , above which the MCFAs become toxic in both cell types, particularly hepatocytes. Concentrations of 1–100 μM C_{11} and C_{12} reduced CN, and this was far more cytotoxic than C_{10} . A 1 mM threshold concentration of C_{10} changed all parameters except PMP on Caco-2 monolayers at both 60 min and 72 h.

We then analysed the data from 1, 8, 24 and 48 h for C_{10} in order to discern patterns with increasing concentrations and exposure times (Fig. 6). The sequence of responses remained consistent across all time-points, with initial increases in NI associated with increases in IC and mitochondrial hyperpolarisation, followed by increased PMP, nuclear fragmentation, and mitochondrial membrane collapse. At all time-points, IC was maintained at high levels before loss of MMP and PMP. Significant reductions in IC only

occurred at or above those concentrations resulting in loss of MMP and significant increases in PMP. In sum, HCA data show that the minimum concentrations of MCFAs needed to induce cellular changes in Caco-2 cells decrease with increasing chain length, indicating an association between increased capacity to increase drug flux and cytotoxicity. Significant changes in IC, PMP and MMP are seen at the same concentrations of MCFAs that promote increased Papp across filter-grown monolayers and it seems that IC changes are the triggering event.

3.6. Comparison of HCA on differentiated filter grown monolayers: C_{10}

HCA and MTT data achieved thus far were on cells grown on plates. Fig. 7 shows representative imaging for Transwell®-grown Caco-2 monolayers grown for 21 days and exposed to 8.5 mM C_{10} or medium. Table 5 compares the data between Caco-2 cells grown on 96 well plates and on Transwells®. The sequence of responses observed for Caco-2 grown on wells was different than those seen for differentiated cells on Transwells®, which displayed a single ‘threshold’ concentration effect across a range of parameters. 8.5 mM C_{10} caused multiple intracellular effects on filter-grown cells, including increases in IC, MMP, NI and PMP. Although this

Table 2
Effect of MCFAs on Caco-2 cell parameters following 60 min exposure.

	C ₈	C ₉	C ₁₀	C ₁₁	C ₁₂
NI	30mM***	7.5mM**	2.5mM**	2.5mM***	2mM***
IC	30mM***	7.5mM*	5mM*	2.5mM*	1mM***
PMP	40mM***	12.5mM***	8.5mM*	2.5mM***	2mM***
NA	75mM***	30mM***	10mM**	13mM*	2mM*
MMP	30mM**	12.5mM**	5mM*	—	2mM**
			20mM**		
CN	40mM***	40mM***	—	—	10mM**

Values are the threshold concentrations that produced a significant change in a parameter compared to untreated controls. White box = increase in parameter value. Grey box = decrease in parameter value. NI = nuclear intensity; IC = intracellular calcium; PMP = plasma membrane potential; NA = nuclear area; MMP = mitochondrial membrane potential; CN = cell number.
**P* < 0.05 compared to controls (*n* = 3).
***P* < 0.01 compared to controls (*n* = 3).
****P* < 0.001 compared to controls (*n* = 3).

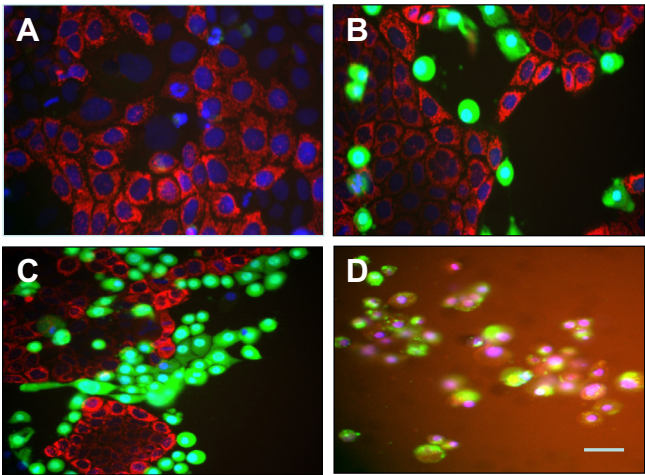


Fig. 5. Representative HCA images of Caco-2 cells after 60 min exposure to C₁₀. (A) Untreated; (B) 5 mM; (C) 8.5 mM; (D) 13 mM. Each fused fluorescent image was acquired at 20× objective magnification and pseudo-coloured accordingly; Hoechst® (nuclear staining; blue), Fluo-4 AM (green; measure of IC), TOTO®-3 iodine (magenta; measure of PMP) and TMRM (red; measure of MMP). Horizontal bar = 20 µm.

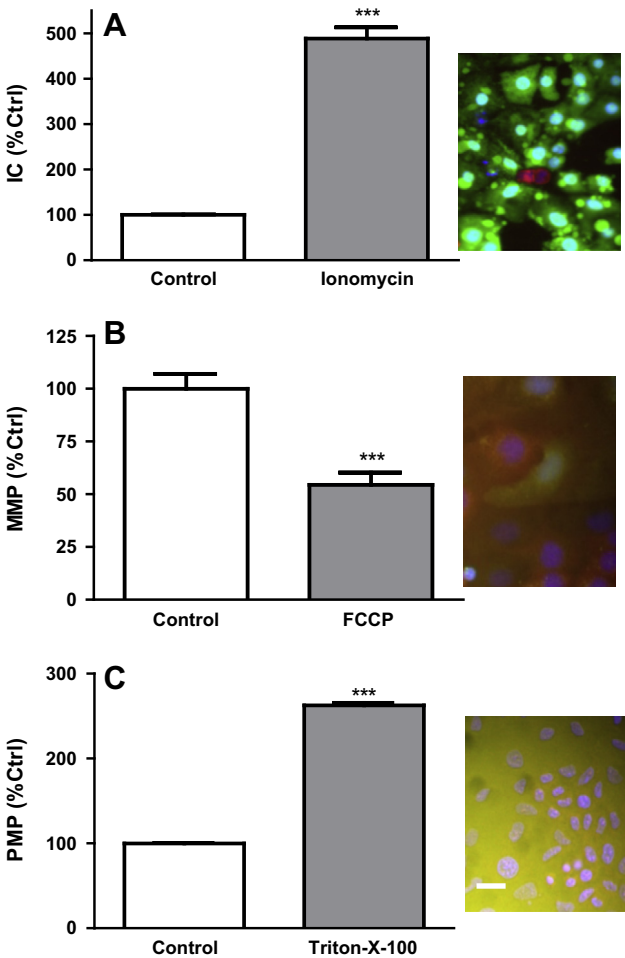


Fig. 4. Positive controls from HCA in Caco-2 cells following 10 min exposure. (A) Ionomycin (20 µM). (B) FCCP (100 µM). (C) Triton-X-100 (0.05% v/v). Horizontal bar = 16 µm. *N* = 3 for each group representing 3 wells in triplicate per plate across 3 different plates. ****P* < 0.001 compared to medium control.

Table 3
Effect of MCFAs on Caco-2 cell parameters following 72 h exposure.

	C ₈	C ₉	C ₁₀	C ₁₁	C ₁₂
NI	5mM***	2.5mM***	1mM***	0.1mM**	0.1mM***
IC	5mM***	2.5mM***	1mM***	0.1mM*	1µM***
	40mM***	15mM***	5mM*	2.5mM*	2mM***
PMP	5mM***	5mM***	2.5mM***	0.5mM***	1mM***
NA	2.5mM***	7.5mM***	1mM***	0.5mM***	2mM**
	10mM*				
MMP	5mM***	2.5mM***	1mM***	0.1mM*	1µM*
	30mM**	10mM**		2.5mM**	1mM**
CN	2.5mM***	2.5mM***	1µM**	0.1mM*	0.1mM***
			1mM***		

Values are minimum concentrations that produced a significant change in a parameter compared to untreated controls. White box = increase in parameter value. Grey box = decrease in parameter value. Code as for Fig. 3.
**P* < 0.05 compared to untreated controls (*n* = 3).
***P* < 0.01 compared to untreated controls (*n* = 3).
****P* < 0.001 compared to untreated controls (*n* = 3).

concentration increased PMP in both construct types, the increase in PMP at 8.5 mM was far more significant in filter-grown cells, likely due to enhanced microvilli, which increases the membrane surface area. Since 8.5 mM C₁₀ was also the minimum concentration required to produce a statistical increase in the *P*_{app} of [¹⁴C]-mannitol, this would associate the PMP increase as a major part of the mechanism of C₁₀-induced permeation enhancement across intestinal epithelia.

4. Discussion

MCFAs are mild anionic surfactants with a hydrophilic carboxyl head group and a single hydrophobic hydrocarbon alkyl tail.

Table 4
Effect of MCFAs on HepG2 cell parameters following 72 h exposure.

	C ₈	C ₉	C ₁₀	C ₁₁	C ₁₂
NI	0.1mM*	0.5mM**	1mM**	10μM**	10μM***
	10mM**	15mM*	8.5mM***	4mM***	2mM***
IC	0.5mM*	1.0mM***	1mM***	10μM**	1mM***
	7.5mM***	7.5mM***	2.5mM**	2.5mM***	1mM***
PMP	5mM*	5mM***	2.5mM***	0.1mM***	10μM**
NA	5mM*	1mM**	1mM***	10μM***	10μM***
MMP	0.1mM**	0.5mM***	1mM***	10μM**	1μM**
	5mM***	2.5mM***			
CN	1μM***	10μM***	1mM***	10μM***	1μM**
	1mM***	2.5mM*			

Values are minimum concentrations that produced a significant change in a parameter compared to untreated controls. White box = increase in parameter value. Grey box = decrease in parameter value. Code as for Fig. 3.

* $P < 0.05$ compared to untreated controls ($n = 3$).

** $P < 0.01$ compared to untreated controls ($n = 3$).

*** $P < 0.001$ compared to untreated controls ($n = 3$).

The amphiphilicity confers at least part of their intestinal permeation enhancing effect by fluidization of intestinal epithelial plasma membranes [23,24]. The CMC of an amphiphilic molecule is the concentration above which the free monomer concentration is at its highest and the surfactant starts to form secondary micelle structures in solution. Increasing the alkyl chain length imparts greater hydrophobicity to fatty acids [25], resulting in lower CMC values (Table 1). MCFAs with low CMC values will therefore have an increased capacity to fluidize biological membranes and to pro-

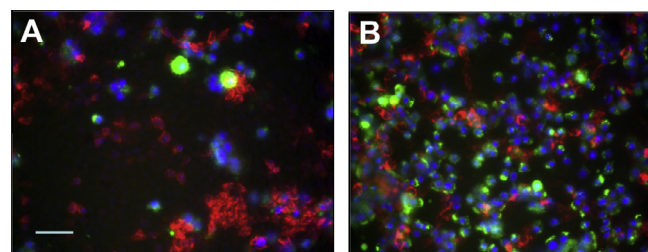


Fig. 7. Representative live-cell images of Caco-2 monolayers grown on Transwells® for 21 days. (A) Untreated monolayers, (B) 60 min treatment of with 8.5 mM C₁₀. Horizontal bar = 30 μm.

mote permeation. This would also be reflected in increased cytotoxicity in relation to C-chain length (MTT data). Use of the sodium salts of MCFAs for studies in aqueous media is an additional variable that further alters physicochemical properties, since conversion of the free fatty acid to the salt increases the hydrophilicity of the carboxyl head group, thereby increasing the hydrophilic-lipophilic balance (HLB), thereby enhancing solubility. Another variable is that the pKa of an MCFA is proportional to alkyl chain length [26]. As the C-chain length increases above C₆, van der Waals interactions between alkyl chains and associated polar head groups reduce intermolecular distances, and the proximity of charged groups to the carboxyl groups stabilises the acid proton, thereby increasing the apparent pKa, hence the reported values for C₈, C₁₀ and C₁₂ are 6.5, 7.0 and 7.5 respectively [26]. As the MCFA salts are weakly ionisable, there is a conversion of non-dissociated carboxylic fatty acid to carboxylate anions and then to sodium salts, as the intestinal pH increases in the small intestine [27]. Across the pH range of 4.5–7.5 range from the small intestine through to colon, and in *in vitro* systems, molecular species will co-exist in varying proportions. Thus, the MCFA physicochemical properties will change according to the many variables encountered in different intestinal regions.

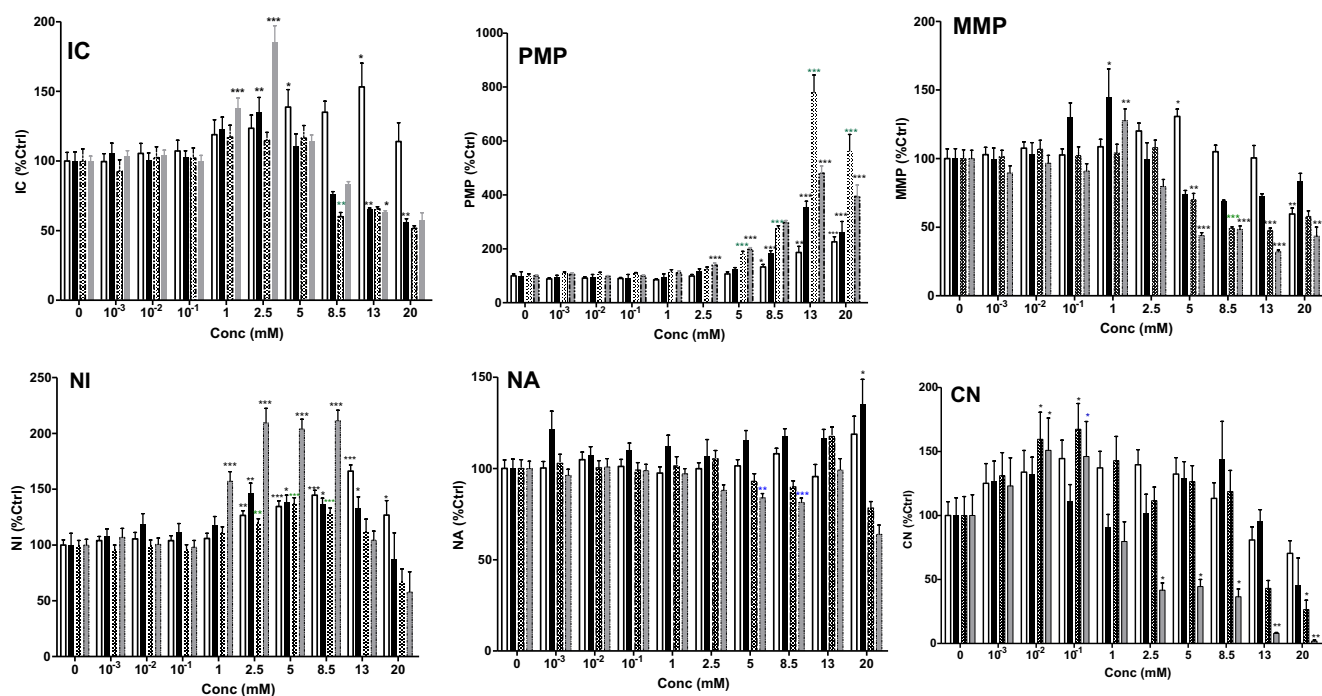


Fig. 6. Caco-2 cell parameter responses on 96 well plates to ascending concentrations of C₁₀ over time. Code: □ 1 h; ■ 8 h; ▨ 24 h; ▩ 48 h. Code: intracellular calcium (IC), plasma membrane potential (PMP), mitochondrial membrane potential (MMP), nuclear intensity (NI), nuclear area (NA), cell number (CN). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ compared to untreated controls ($n = 3$).

Table 5

Comparison of effect of C₁₀ on Caco-2 cell parameters following 60 min exposure on Transwells® versus 96 well plates.

	96 well plates	Transwells®
NI	2.5mM**	8.5mM***
IC	5.0mM*	8.5mM*
PMP	8.5mM*	8.5mM***
NA	10.0mM**	5.0mM*
MMP	5mM*	8.5mM***
	20mM**	
CN	–	15mM

Values are the threshold concentrations that produced a significant change in a parameter compared to untreated controls. White box = increase in parameter value. Grey box = decrease in parameter value. NI = nuclear intensity; IC = intracellular calcium; PMP = plasma membrane potential; NA = nuclear area; MMP = mitochondrial membrane potential; CN = cell number.

*P < 0.05 compared to controls (n = 3).

**P < 0.01 compared to controls (n = 3).

***P < 0.001 compared to controls (n = 3).

Traube's rule [28] predicts that for each hydrocarbon added to the alkyl chain, the concentration of MCFA required to significantly increase cell membrane fluidity should decrease approximately three fold. The EC₅₀ concentrations of the MCFA series required to increase the Papp of [¹⁴C]-mannitol across monolayers followed this prediction. The CMC values also followed Traube's rule: the experimentally determined value for C₈ was 262 mM, and this predicted the values of 26.1 and 2.8 mM for C₁₀ and C₁₂ respectively. The HCA data generated for the series following 1 h exposure to Caco-2 cells also show a correlation between chain length and the minimum concentration required to increase IC and PMP, the two cell parameters most associated with increased permeability at similar concentrations. HCA assessment in HepG2 and Caco-2 cell therefore revealed sub-lethal cytotoxicity of the order C₁₂ > C₁₁ > C₁₀ > C₉ = C₈.

Lindmark et al. also showed that the concentration required to significantly increase [¹⁴C]-mannitol transepithelial permeability across Caco-2 monolayers decreased with increasing chain length for even numbered MCFA from C₆–C₁₂ [14]. However, in that study no correlations were apparent between MCFA chain length and effects on the cytoskeleton or cell morphology. While C₈ demonstrated greatest effect on apical membrane permeability using propidium iodide as a marker, C₁₀ and C₁₂ showed no effect. The authors suggested that while the initial interaction of MCFAs with the apical membrane of the cell monolayer was a CMC-dependent response, study of intracellular mechanisms of action did not show correlation with chain length, unlike the events revealed here by HCA. A subsequent Caco-2 study examining effects of C₉–C₁₄ demonstrated a direct correlation between MCFA chain length and permeation enhancement of penicillin across monolayers and cytotoxicity in cells [29], but they found that C₁₃ and C₁₄ were the least toxic of the fatty acids tested. There are differing mechanisms by which medium and long chain fatty acids are processed following interaction with the enterocyte membrane. Differences in fatty acid solubility and internalisation from the membrane surface into the aqueous cytosolic environment are likely to occur in the series [30]. For example, C₈–C₁₂ enter the mitochondria independently of the carnitine transport system and are oxidised or

pass directly across the basolateral membrane. In contrast, longer chain fatty acids including C₁₃ and C₁₄ may be diverted into chylomicrons *en route* to the lymphatic system [31], thereby reducing intracellular effects and cytotoxicity.

HCA revealed the fundamental role played by disruption of Ca²⁺ homeostasis that may ultimately lead to the irreversible mitochondrial injury seen in the MTT assay. MCFA perturbation of cells induces typical early stress responses including chromatin condensation (reflected in increased NI), and increased IC. Concentrations below the CMC do not cause an increase in permeability sufficient to allow appreciable monolayer drug permeation, but such concentrations are still likely to be sufficient to disrupt PLC and cause release of Ca²⁺ from the endoplasmic reticulum. The mitochondrial membrane potential is generated by the release of protons from oxidised NADH and by active transport across the inner mitochondrial membrane. A high affinity Ca²⁺-selective ion conductance channel on the inner mitochondrial membrane ensures that only Ca²⁺ can dissipate the MMP [32]. As the mitochondria sequester excess IC, they become hyperpolarised, increase ATP production, which contributes to the active Ca²⁺ transport out of the cell by plasma membrane Ca²⁺ATP-ase (PMAC). Increased ATP production also produces more oxygen which is reduced to water, but there is also disruption to the electron transport chain during energy production with increased leakage of free electrons leading to production of reactive oxygen species, which oxidise amino acids and polyunsaturated fatty acids in lipid membranes and cause damage to DNA. If cell perturbation is relieved before the irreversible phase of cell damage is induced, then the mitochondria release Ca²⁺ back to the cytoplasm, with the MMP returning to normal. Ca²⁺ overload in mitochondria however leads to collapse and loss of membrane potential through opening of permeability transition pores (PTP), a multi-ion channel. Sequestered Ca²⁺ is released back into the cytoplasm, resulting in irreversible cellular damage [33]. The opening of PTP releases pro-apoptotic proteins including cytochrome c from the mitochondrial intermembrane space, activates capases, leading to chromatin condensation (increased NI), nuclear fragmentation (decreased NA) and cell death [34]. The maintenance of high IC detected by HCA following MCFA exposure may be the result of PMAC disruption via Ca²⁺ activated calpain [35], or by MCFA-induced perturbation of ER membranes and mitochondrial collapse.

Many of the MCFA induced cellular responses revealed by HCA are therefore mediated by disruption of intracellular Ca²⁺ homeostasis and this is the over-arching link for MCFA to the paracellular route. PLC activation of the IP₃ pathway increases IC concentrations leading to Ca²⁺-calmodulin mediated disruption of F-actin, contraction of the peri-junctional actinomyosin ring [13], myosin light chain kinase inhibition [36], and the removal of tricellulin and claudin 5 from TJs [15]; these are proposed TJ mechanisms of action of C₁₀ in intestinal epithelial models *in vitro*. We found that the concentrations required to increase the mannitol Papp across Caco-2 monolayers for each MCFA were at or above the concentrations causing significant increase in PMP. The increase in PMP is therefore likely to be a result of direct MCFA perturbation and/or Ca²⁺ mediated alterations to membrane morphology. In any event, it appears that elevated IC and increased PMP are two of the key initiating events that precede overall effects of MCFA on transepithelial permeation.

5. Conclusions

A HCA cytotoxicity protocol was modified for the study of MCFA cellular effect on Caco-2 cells and demonstrated SAR with regards to lipophilicity, toxicity, and permeation enhancement. MCFA lipophilicity increases with increasing alkyl chain length, increasing

extracellular membrane perturbation and intracellular Ca^{2+} homeostasis disruption. Both effects ultimately lead to alterations in cell morphology and increases in drug permeation. *In vitro*, MCFA monomers sufficiently disrupt cell monolayer barrier function with permeation enhancement observed at concentrations well below the CMC. The intracellular mechanism of action is highly complex, concentration-dependent and model-dependent: it involves phospholipase C-mediated elevation of intracellular Ca^{2+} , subsequent regulation of tight junction proteins and detergent-induced membrane fluidization. Finally, we note that the correlations were limited to *in vitro* assays of intestinal monolayers; it will be interesting to discover if there are valid correlations with respect to pharmacodynamic read-outs from *ex-vivo* and *in situ* intestinal models.

Conflict of Interest Statement

The co-sponsor, Merrion Pharmaceuticals is an oral drug delivery company, but it was not involved in the research, except to approve the publication according to the terms of its contract with UCD.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ejpb.2014.10.008>.

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